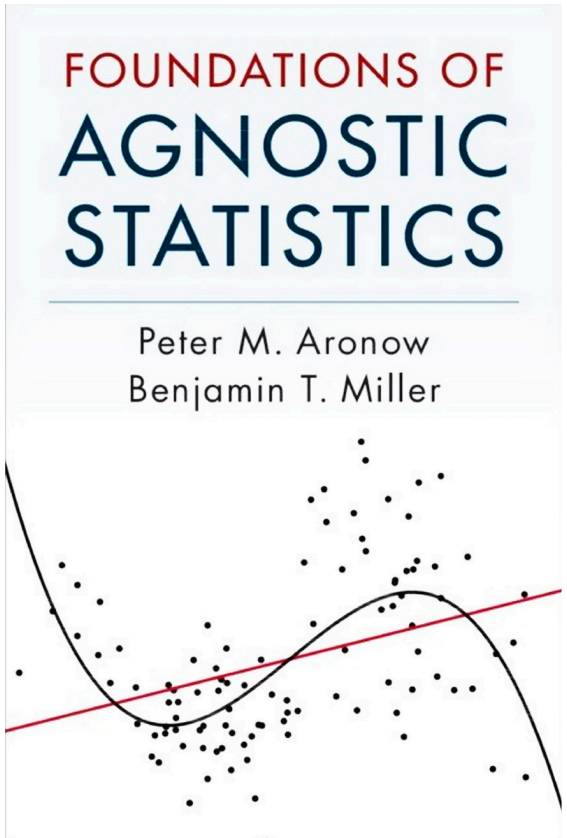
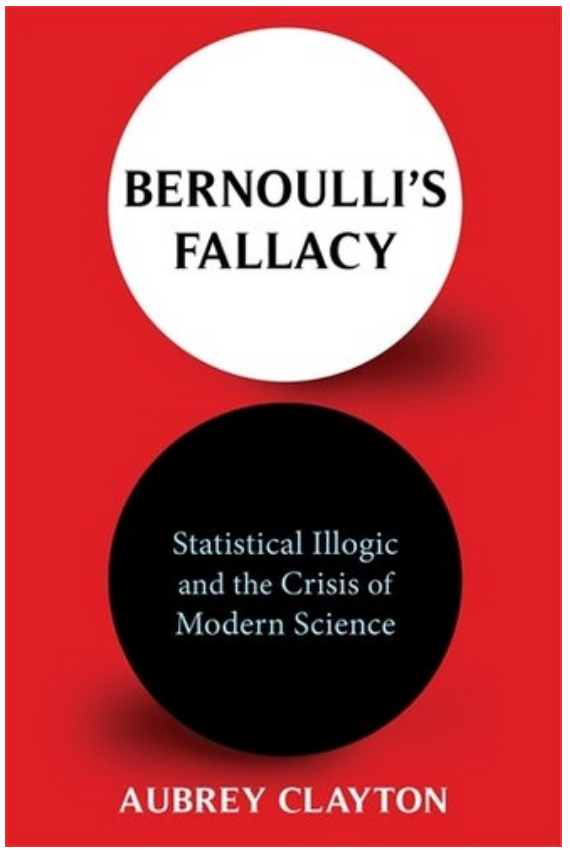
Probability Theory

X610 · Session 1 · *FoAS* Chapter 1

Course texts





Aronow & Miller — *FoAS*

Clayton — *BF*

Today

- Probability spaces, events, and the axioms of probability
- Joint and conditional probability, independence, and Bayes' rule
- Random variables, support, and distribution functions
- Joint, marginal, and conditional distributions of multiple random variables

Reading: *FoAS* Ch. 1; *Bernoulli's Fallacy* Intro + Ch. 1

Probability foundations

Notation (1)

Notation	Definition and usage
$\mathbb{R}$	The set of all real numbers — all the points on a continuous number line.
$\in$	Set membership. $s \in A$ (read "s is an element of A") denotes that the object s is an element of the set A.
$\subseteq$	Subset. $A \subseteq B$ means B contains every element of A: $A \subseteq B \iff \forall s \in A, s \in B$. For example, $\{2, 3, 5\} \subseteq \{1, 2, 3, 4, 5\}$.
$\forall$	For all. States that all elements of a particular set satisfy a given expression. For example, $\forall x \in \{1, 2, 3, 4, 5\}, 2x \leq 10$.
A^C	The complement of a set: all elements of the universal set U that are not in A. Formally, $A^C = \{s \in U : s \notin A\}$.
$A \setminus B$	Set subtraction: all elements in A except any also in B. Formally, $A \setminus B = \{s \in A : s \notin B\}$.
$\emptyset$	The empty set — the set containing no elements.

Notation (2)

Notation	Definition and usage
$\cup$	Union. $A \cup B$ contains all elements in either A or B (or both). For example, $\{1, 2, 5\} \cup \{2, 3\} = \{1, 2, 3, 5\}$.
$\cap$	Intersection. $A \cap B$ contains all elements in both A and B. For example, $\{1, 2, 5, 6\} \cap \{2, 3, 6\} = \{2, 6\}$.
$\lvert A \rvert$	The cardinality (size) of a set. For finite sets, the number of elements: $\lvert\{1, 2, 3, 5, 6, 8\}\rvert = 6$.
$\mathcal{P}(A)$	The power set of A: the set of all subsets of A.

Notation (3)

Notation	Definition and usage
$\sum$	Summation of a sequence. For example, $\sum_{j=1}^{4} j^2 = 1 + 4 + 9 + 16 = 30$.
$f : S \to T$	A function f from S to T. S is the domain; T is the codomain, containing all possible values of $f(s)$.
$\iff$	If and only if. Two statements are logically equivalent; each implies the other. For example, $2x = 10 \iff x = 5$.
$\square$	Quod erat demonstrandum (QED). Denotes that a proof is complete.
$\triangle$	End of example (*FoAS*'s nonstandard delimiter).

Random generative processes

- A **random generative process** is a mechanism that selects an outcome from among multiple possible outcomes
- A single instance of selecting an outcome is a **draw** or **realization**
- Examples: flipping a coin, rolling a die, drawing a ball from an urn, selecting a person at random from a group

Probability is a *model*

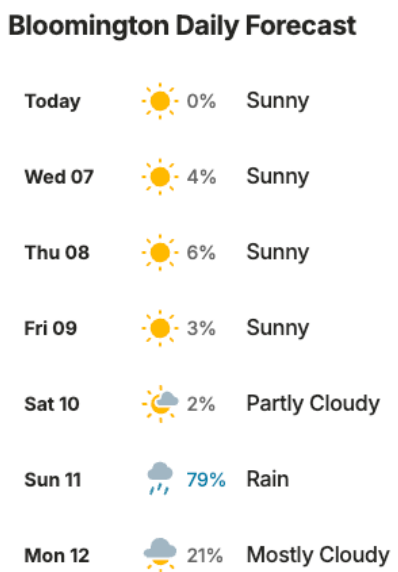
> Probability theory is a model, which is an approximation of reality. Everyday macrophysical processes are not actually characterized by fundamental randomness. … The mathematical construct of randomness is, therefore, a modeling assumption, not necessarily a fundamental feature of reality. —*FoAS*, pp. 4–5

> All models are wrong, but some are useful. —George Box

> Truth is much too complicated to allow anything but approximations. —John von Neumann

Weather forecasts

What do these probabilities mean?



Frequentist probability

- *FoAS* adopts the **frequentist** interpretation:

> "The probability of an event describes the proportion of times that event can be expected to occur among many realizations of a random generative process. … It is the long-run average or limiting value of the frequency of observing event A among repeated realizations of the generative process." (*FoAS*, p. 4)

- *Bernoulli's Fallacy* challenges this interpretation

Python setup

Every code example in this course assumes this preamble:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

rng = np.random.default_rng(610)
```

- `rng` is our *seeded* random number generator: everyone's simulations give identical results

Long-run frequency, simulated

```
flips = rng.integers(0, 2, size=100_000)   # 1 = heads, 0 = tails
running_freq = np.cumsum(flips) / np.arange(1, flips.size + 1)

print(running_freq[[9, 99, 999, 9_999, 99_999]])
```

```
[0.4     0.38    0.48    0.4959  0.49908]
```

- The relative frequency of heads stabilizes near $\frac{1}{2}$ as draws accumulate
- This is the frequentist notion of probability

Probability space

Component	Symbol	What it is
Sample space	Ω	Set of all possible outcomes ω
Event space	S	Set of subsets of Ω (the events)
Probability measure	P	Function assigning a probability to every event

- Die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$
  - "roll an even number" is the event $A = \{2, 4, 6\}$

Event space

Definition 1.1.1 — Event Space

A set S of subsets of Ω is an **event space** if it satisfies the following:

- Nonempty: $S \neq \emptyset$.
- Closed under complements: if $A \in S$, then $A^C \in S$.
- Closed under countable unions: if $A_1, A_2, A_3, \ldots \in S$, then $A_1 \cup A_2 \cup A_3 \cup \ldots \in S$.

- Why:
  - if A has a probability, "not A" must also;
  - if "A and B" has a probability, "A or B" must also

The Kolmogorov axioms

Definition 1.1.2 — Kolmogorov Axioms

(Ω, S, P) is a **probability space** if it satisfies the following:

- Non-negativity: $\forall A \in S, P(A) \geq 0$, where $P(A)$ is finite and real.
- Unitarity: $P(\Omega) = 1$.
- Countable additivity: if $A_1, A_2, A_3, \ldots \in S$ are pairwise disjoint, then

$$P(A_1 \cup A_2 \cup A_3 \cup \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots = \sum_i P(A_i).$$

The Kolmogorov axioms, in words

1. No event has negative probability
2. *Something* in Ω has to happen
3. For mutually exclusive events, probabilities add

- Everything else in probability theory is *derived* from these three axioms

A fair coin flip

Let H represent the outcome "heads" and T represent the outcome "tails." Let $\Omega = \{H, T\}$ and $S = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$. Define the probability measure as:

$$P(A) = \frac{1}{2}|A|, \quad \forall A \in S.$$

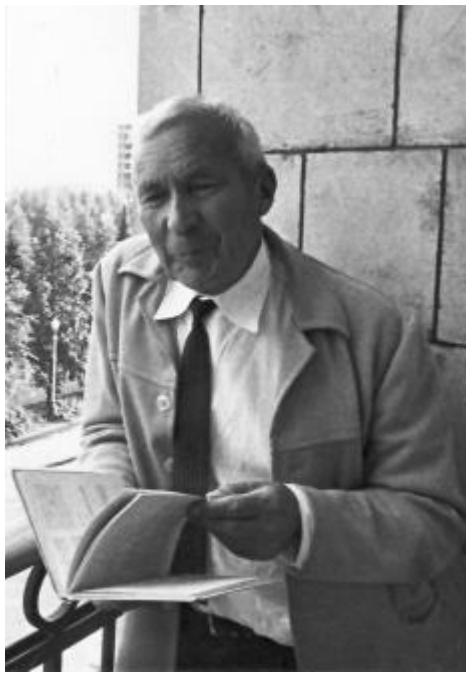
This means that:

- $P(\emptyset) = \frac{1}{2}|\emptyset| = \frac{1}{2} \cdot 0 = 0$. The probability of nothing happening is zero.
- $P(\{H\}) = \frac{1}{2}|\{H\}| = \frac{1}{2} \cdot 1 = \frac{1}{2}$. The probability of getting heads is $\frac{1}{2}$.
- $P(\{T\}) = \frac{1}{2}|\{T\}| = \frac{1}{2} \cdot 1 = \frac{1}{2}$. The probability of getting tails is $\frac{1}{2}$.
- $P(\{H, T\}) = \frac{1}{2}|\{H, T\}| = \frac{1}{2} \cdot 2 = 1$. The probability of getting either heads or tails is one.

Since S is a proper event space and that P satisfies the Kolmogorov axioms, (Ω, S, P) is a probability space.

Andrey Kolmogorov (1903–1987)

- Soviet mathematician, professor at Moscow State University, and one of the most influential mathematicians of the 20th century
- His 1933 monograph *Foundations of the Theory of Probability* put probability on a rigorous footing by grounding it in measure theory
- Before Kolmogorov, probability was a collection of useful techniques with disputed foundations; after him, it was a branch of mathematics
- He made fundamental contributions beyond probability: turbulence, dynamical systems, and algorithmic complexity
- Famously devoted to teaching, he also wrote school textbooks and mentored dozens of doctoral students



Basic properties of probability

Theorem 1.1.4 — Basic Properties of Probability

Let (Ω, S, P) be a probability space. Then, $\forall A, B \in S$:

- Monotonicity: if $A \subseteq B$, then $P(A) \leq P(B)$.
- Subtraction rule: if $A \subseteq B$, then $P(B \setminus A) = P(B) - P(A)$.
- Zero probability of the empty set: $P(\emptyset) = 0$.
- Probability bounds: $0 \leq P(A) \leq 1$.
- Complement rule: $P(A^C) = 1 - P(A)$.

Joint probability

Definition 1.1.5 — Joint Probability

For $A, B \in S$, the **joint probability** of A and B is $P(A \cap B)$.

Example 1.1.6 — A Fair Die Roll

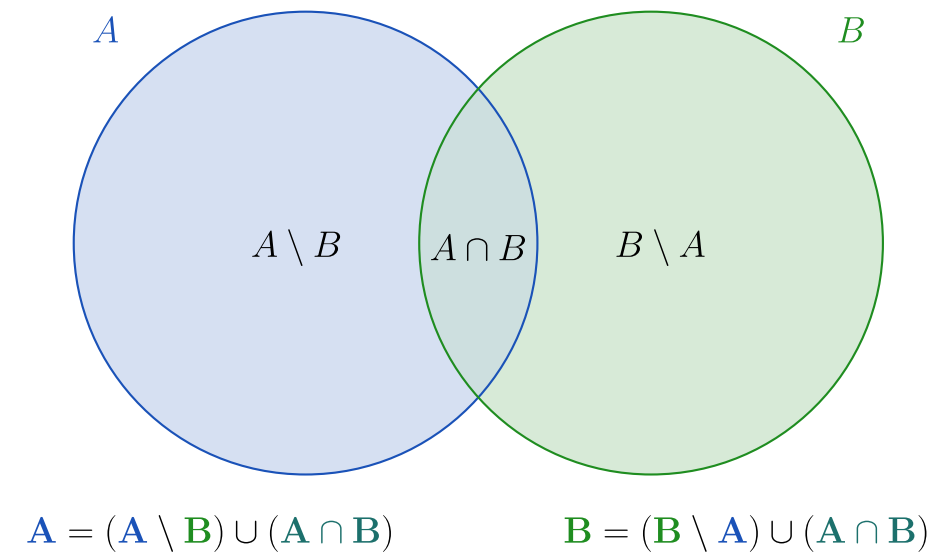
$\Omega = \{1, \dots, 6\}$, $P(A) = \frac{1}{6}|A|$. Let $A = \{4, 5, 6\}$ ("at least 4") and $B = \{2, 4, 6\}$ ("even"). Then

$$P(A \cap B) = P(\{4, 6\}) = \frac{2}{6} = \frac{1}{3}.$$

Addition rule

Theorem 1.1.7 — Addition Rule

For $A, B \in S$, $\quad P(A \cup B) = P(A) + P(B) - P(A \cap B).$



- Adding $P(A)$ and $P(B)$ double-counts the overlap $A \cap B$
- Disjoint events: reduces to $P(A \cup B) = P(A) + P(B)$

Joint probability, simulated

```
rolls = rng.integers(1, 7, size=100_000)

A = rolls >= 4          # Boolean value that is True if roll>=4
B = rolls % 2 == 0      # Boolean value that is True if roll is even

print('P(A∩B) ≈', (A & B).mean())            # 1/3
print('P(A∪B) ≈', (A | B).mean())            # 1/2 + 1/2 - 1/3 = 2/3
print('check  :', A.mean() + B.mean() - (A & B).mean())
```

```
P(A∩B) ≈ 0.33009
P(A∪B) ≈ 0.66688
check  : 0.66688
```

- Boolean arrays make event logic transparent: `&` is intersection, `|` is union, `.mean()` is relative frequency

Conditional probability

Definition 1.1.8 — Conditional Probability

For $A, B \in S$ with $P(B) > 0$, the **conditional probability** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- Conditioning = restricting attention to the draws where B occurred

Multiplicative law

Theorem 1.1.9 — Multiplicative Law of Probability

For $A, B \in S$ with $P(B) > 0$,

$$P(A \mid B)\, P(B) = P(A \cap B).$$

- Just a rearrangement of Definition 1.1.8 — but the rearranged form is how joint probabilities are usually *computed*
- In a multi-stage process, the conditional probabilities are what we know directly: each stage's probabilities given the earlier stages
- The multiplicative law chains them into the probability that the whole sequence occurs. Example: Flip a coin, then roll a die…

$$P(\text{heads} \cap \text{roll } 3) = P(\text{roll } 3 \mid \text{heads})\, P(\text{heads})$$

Conditioning, simulated

```
print('P(A|B) ≈', A[B].mean())      # restrict to draws where B occurred
print('exact  :', (2/6) / (3/6))    # P(AnB)/P(B) = 2/3
```

```
P(A|B) ≈ 0.661980587197176 3
exact  : 0.6666666666666666
```

- `A[B]` keeps only the realizations in which B occurred — the simulation analog of conditioning
  - This "Boolean masking" is a common idiom in both Python and R

Bayes' rule

Theorem 1.1.10 — Bayes' Rule

For $A, B \in S$ with $P(A) > 0$ and $P(B) > 0$,

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}.$$

- Reverses the direction of conditioning: relates $P(A \mid B)$ to $P(B \mid A)$

Where our books disagree

- *FoAS*: “Although we will not be making use of Bayes’ Rule in this book, we would be remiss to neglect it entirely.”
- *BF*: this reversal is kind of the *whole point* — inference means getting from $P(\text{data} \mid \text{hypothesis})$ to $P(\text{hypothesis} \mid \text{data})$

A two-stage process

Example 1.1.11 — Flipping a Coin and Rolling a Die

Flip a fair coin. Heads → roll a fair **four**-sided die; tails → roll a fair **six**-sided die. Let A = heads, B = observing 3.

$$P(A \cap B) = P(B \mid A)P(A) = \tfrac{1}{4} \cdot \tfrac{1}{2} = \tfrac{1}{8}, \qquad P(A^C \cap B) = \tfrac{1}{6} \cdot \tfrac{1}{2} = \tfrac{1}{12}.$$

$$P(B) = P(A \cap B) + P(A^C \cap B) = \tfrac{1}{8} + \tfrac{1}{12} = \tfrac{5}{24}.$$

By Bayes' rule:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{5}{24}} = \frac{3}{5}.$$

- Seeing a 3 makes heads *more* likely than the unconditional $\frac{1}{2}$ — a 3 is easier to get on a four-sided die

The same process, simulated

```
n = 100_000
heads = rng.random(n) < 0.5
die = np.where(heads,
               rng.integers(1, 5, size=n),    # 4-sided if heads
               rng.integers(1, 7, size=n))    # 6-sided if tails

three = die == 3
print('P(heads | rolled 3) ≈', heads[three].mean())
```

```
P(heads | rolled 3) ≈ 0.598079428654152 4
```

- The simulation models Bayes' rule: among draws where B occurred, count how often A also did

Partitions

Definition 1.1.12 — Partition

If $A_1, A_2, A_3, \ldots \in S$ are nonempty and pairwise disjoint, and $\Omega = A_1 \cup A_2 \cup A_3 \cup \ldots$, then $\{A_1, A_2, A_3, \ldots\}$ is a **partition** of Ω.

- A partition divides Ω into mutually exclusive, exhaustive "bins": exactly one A_i occurs on any draw

Law of Total Probability

Theorem 1.1.13 — Law of Total Probability

If $\{A_1, A_2, A_3, \ldots\}$ is a partition of Ω and $B \in S$, then

$$P(B) = \sum_i P(B \cap A_i).$$

If we also have $P(A_i) > 0$ for $i = 1, 2, 3, \ldots$, then this can also be stated as

$$P(B) = \sum_i P(B \mid A_i)\, P(A_i).$$

- $P(B)$ is a weighted average of conditional probabilities, weighted by the $P(A_i)$

Bayes' rule, again

Theorem 1.1.14 — Alternative Forms of Bayes' Rule

If $\{A_1, A_2, A_3, \ldots\}$ is a partition of Ω with $P(A_i) > 0$ for $i = 1, 2, 3, \ldots$, and $B \in S$ with $P(B) > 0$, then

$$P(A_j \mid B) = \frac{P(B \mid A_j)\, P(A_j)}{\sum_i P(B \mid A_i)\, P(A_i)}.$$

Base rates matter

A disease affects 1% of a population. A test has:

- $P(+ \mid \text{sick}) = 0.95$ (*sensitivity* is 95%)
- $P(- \mid \text{healthy}) = 0.95$ (*specificity* is 95%)

You test positive. 🤒

What is the probability that you have the disease?

- Write down your answer before we simulate it

Base rates matter: the answer

```
n = 1_000_000
sick = rng.random(n) < 0.01
positive = np.where(sick, rng.random(n) < 0.95, rng.random(n) < 0.05)

print('P(sick | +) ≈', sick[positive].mean())
```

```
P(sick | +) ≈ 0.16084212669222678
```

- Roughly **16%**, not 95% — the prior $P(\text{sick}) = 0.01$ dominates
- This is Clayton's *base rate neglect*: $P(+ \mid \text{sick})$ is not $P(\text{sick} \mid +)$

Independence of events

Definition 1.1.15 — Independence of Events

Events $A, B \in S$ are **independent** if $P(A \cap B) = P(A)\,P(B)$.

Theorem 1.1.16 — Conditional Probability and Independence

For $A, B \in S$ with $P(B) > 0$, A and B are independent if and only if

$$P(A \mid B) = P(A).$$

- Independence: knowing whether B occurred tells you *nothing* about A
- Coin-and-die example: $P(A \mid B) = \frac{3}{5} \neq \frac{1}{2} = P(A)$

Random variables and probability distributions

Notation (4)

Notation	Definition and usage
$\exists$	There exists. States that at least one element of a set satisfies an expression. For example, $\exists x \in \{1, 2, 3, 4, 5\}$ such that $x^2 = 16$.
$\mathbb{N}$	The set of all natural numbers (positive integers): $\mathbb{N} = \{1, 2, 3, ...\}$.
$\mathbb{Z}$	The set of all integers: $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$.
$\mathbb{R}^n$	The real coordinate space of dimension n: all vectors of length n with real entries.
$f(A)$	The image of the set A under f: all values f can take on when applied to an element of A. Note that $f(S)$ is the range of f — e.g., $X(\Omega)$, the range of a random variable.
$g \circ f$	Composition of two functions: first apply f, then g: $(g \circ f)(s) = g(f(s))$.

Random variables

Definition 1.2.1 — Random Variable

A random variable is a function $X : \Omega \to \mathbb{R}$ such that, $\forall r \in \mathbb{R}$,

$$\{\omega \in \Omega : X(\omega) \leq r\} \in S.$$

- Maps each state of the world to a real number
- "Neither random nor a variable" — it's a function
  - The randomness comes from the underlying generative process, and the variable is the output of the function

Random variables

- Some notation:
  - X is the random variable and x is a particular draw
  - $\{X = 1\}, \{X \geq 0\}$ are events in terms of the r.v.
  - $\Pr[X = 1]$ is the probability that a draw from the r.v. is 1
- So we can say things like $\Pr[X \leq x]$

Discrete random variables

Definition 1.2.4 — Discrete Random Variable

A random variable X is discrete if its range, $X(\Omega)$, is a countable set.

- Can be finite or countably infinite

Definition 1.2.5 — Probability Mass Function (PMF)

For a discrete random variable X, the probability mass function of X is

$$f(x) = \Pr[X = x], \ \forall x \in \mathbb{R}.$$

The Bernoulli random variable

Example 1.2.7 — A Biased Coin Flip (Bernoulli Distribution)

Let $X = 0$ if the coin comes up heads and $X = 1$ if tails, with $\Pr[X = 1] = p$. Then

$$f(x) = \begin{cases} 1 - p & : \ x = 0 \\ p & : \ x = 1 \end{cases}$$

$f(x) = 0$ for all other $x \in \mathbb{R}$.

- The simplest nontrivial random variable
- A whole family tree of related distributions grows out of this example

Properties of PMFs

Theorem 1.2.9 — Properties of PMFs

For a discrete random variable X with PMF f,

- $\forall x \in \mathbb{R},\ f(x) \geq 0$,
- $\sum_{x \in X(\Omega)} f(x) = 1$.

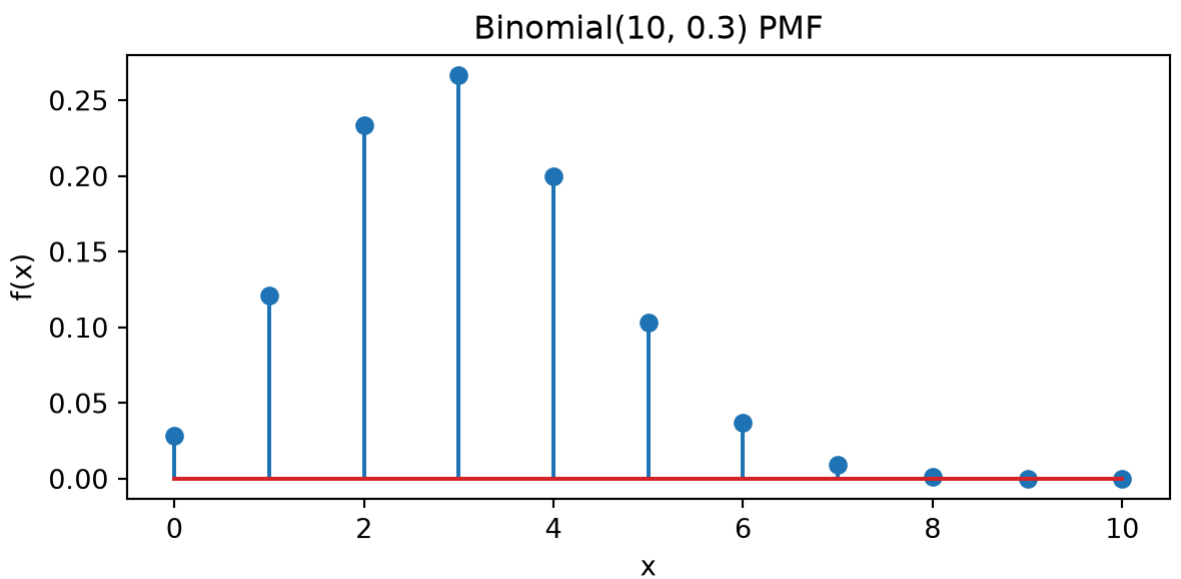
- Probabilities of events: sum the PMF over relevant values
- Die roll example:

$$\Pr[X \geq 3] = \sum_{x=3}^{6} f(x) = \frac{2}{3}$$

A PMF in code

```
X = stats.binom(n=10, p=0.3)     # the r.v.
x = np.arange(11)                # x = [0, 1, 2, ..., 10]

fig, ax = plt.subplots(figsize=(7, 3))
ax.stem(x, X.pmf(x))             # plot (x,y)
ax.set(xlabel='x', ylabel='f(x)', title='Binomial(10, 0.3) PMF')
plt.show()
```



Cumulative distribution functions

Definition 1.2.11 — Cumulative Distribution Function (CDF)

For a random variable X, the cumulative distribution function of X is

$$F(x) = \Pr[X \leq x], \ \forall x \in \mathbb{R}.$$

- *Every* random variable has a CDF — discrete, continuous, or otherwise
- The CDF tells us everything about the distribution of X

Properties of CDFs

Theorem 1.2.12 — Properties of CDFs

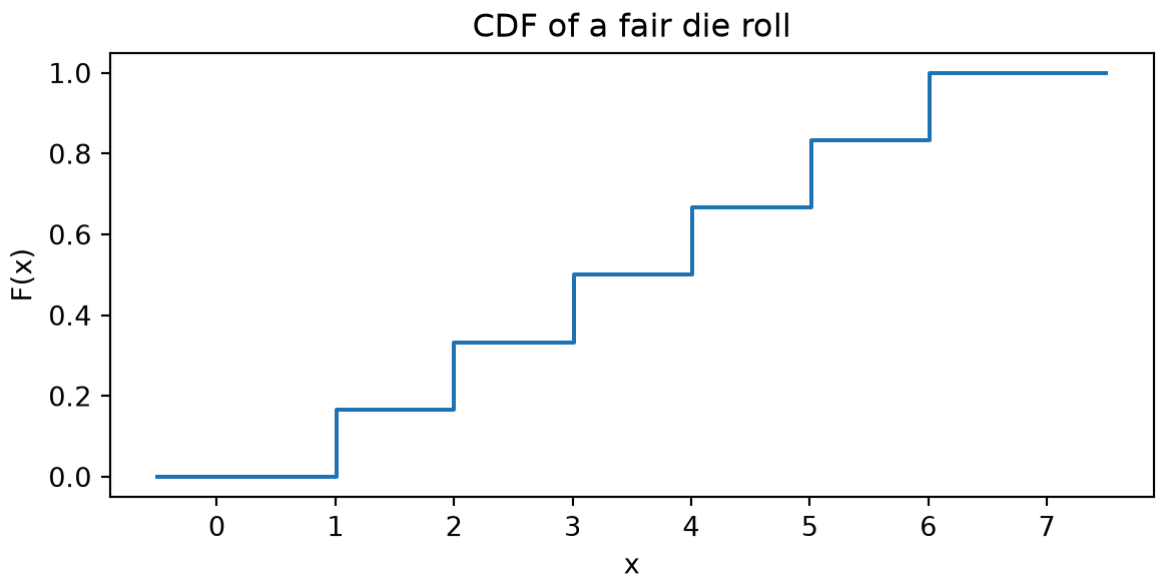
For a random variable X with CDF F,

- F is nondecreasing: $\forall x_1, x_2 \in \mathbb{R}$, if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.
- $\lim_{x \to -\infty} F(x) = 0$.
- $\lim_{x \to \infty} F(x) = 1$.
- $\forall x \in \mathbb{R},\ 1 - F(x) = \Pr[X > x]$.

- For a discrete RV, the CDF is a *step function*:
  - flat, then jumps by $f(x)$

The die-roll CDF

```
1 die = stats.randint(1, 7)          # create the r.v.
2 x = np.linspace(-0.5, 7.5, 500)    # 500 points between -0.5 and 7.5
3
4 fig, ax = plt.subplots(figsize=(7, 3))
5 ax.step(x, die.cdf(x), where='post')    # create the step function
6 ax.set(xlabel='x', ylabel='F(x)', title='CDF of a fair die roll')
7 plt.show()
```



- $F(1.5) = \frac{1}{6}$
- $F(6) = 1$ and $F(x) = 1$ for all $x > 6$

A question

Suppose X is a person's height, measured with unlimited precision.

What is $\Pr[X = 70]$?

- Commit to an answer

- What would your answer imply about the sum of probabilities over all possible heights?

Continuous random variables

Definition 1.2.14 — Continuous Random Variable

A random variable X is continuous if there exists a non-negative function $f : \mathbb{R} \to \mathbb{R}$ such that the CDF of X is

$$F(x) = \Pr[X \leq x] = \int_{-\infty}^{x} f(u)\, du, \ \forall x \in \mathbb{R}.$$

Definition 1.2.15 — Probability Density Function (PDF)

For a continuous random variable X with CDF F, the PDF of X is

$$f(x) = \left. \frac{dF(u)}{du} \right|_{u=x}, \ \forall x \in \mathbb{R}.$$

The answer: zero

From Theorem 1.2.17 (*Event Probabilities for Continuous Random Variables*):

- $\forall x \in \mathbb{R},\ \Pr[X = x] = 0$
  - Follows from $\Pr[a \leq X \leq b] = F(b) - F(a)$
  - Strict vs. weak inequalities make no difference for a continuous r.v.

Properties of PDFs

Theorem 1.2.16 — Properties of PDFs

For a continuous random variable X with PDF f,

- $\forall x \in \mathbb{R},\ f(x) \geq 0.$
- $\int_{-\infty}^{\infty} f(x)\,dx = 1.$

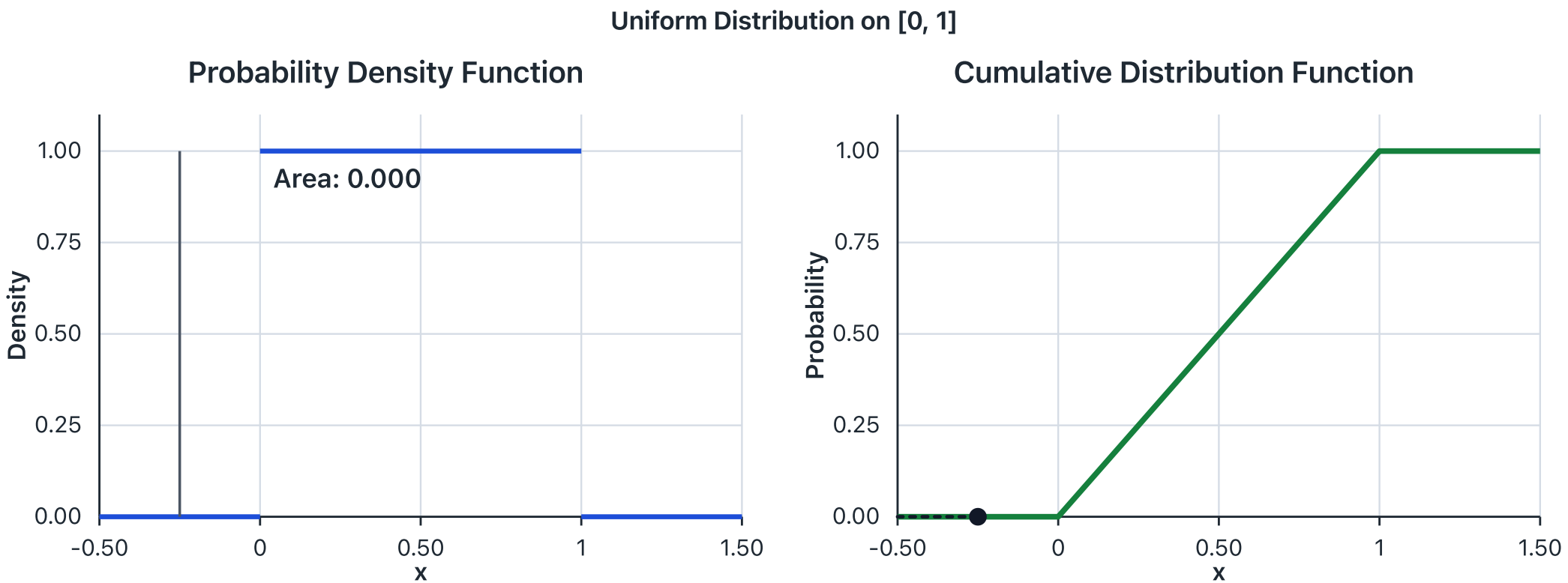
- Same idea as PMF properties, with integrals replacing sums
- The PDF is *density*, not probability: $f(x)$ can exceed 1
  - only areas are probabilities

The standard uniform

Example 1.2.18 — Standard Uniform Distribution

If $X \sim U(0, 1)$, all values in $[0, 1]$ are equally likely: $f(x) = 1$ for $0 \leq x \leq 1$ (0 otherwise), so $F(x) = x$ on $[0, 1]$.

Select a value for x: **-0.25**



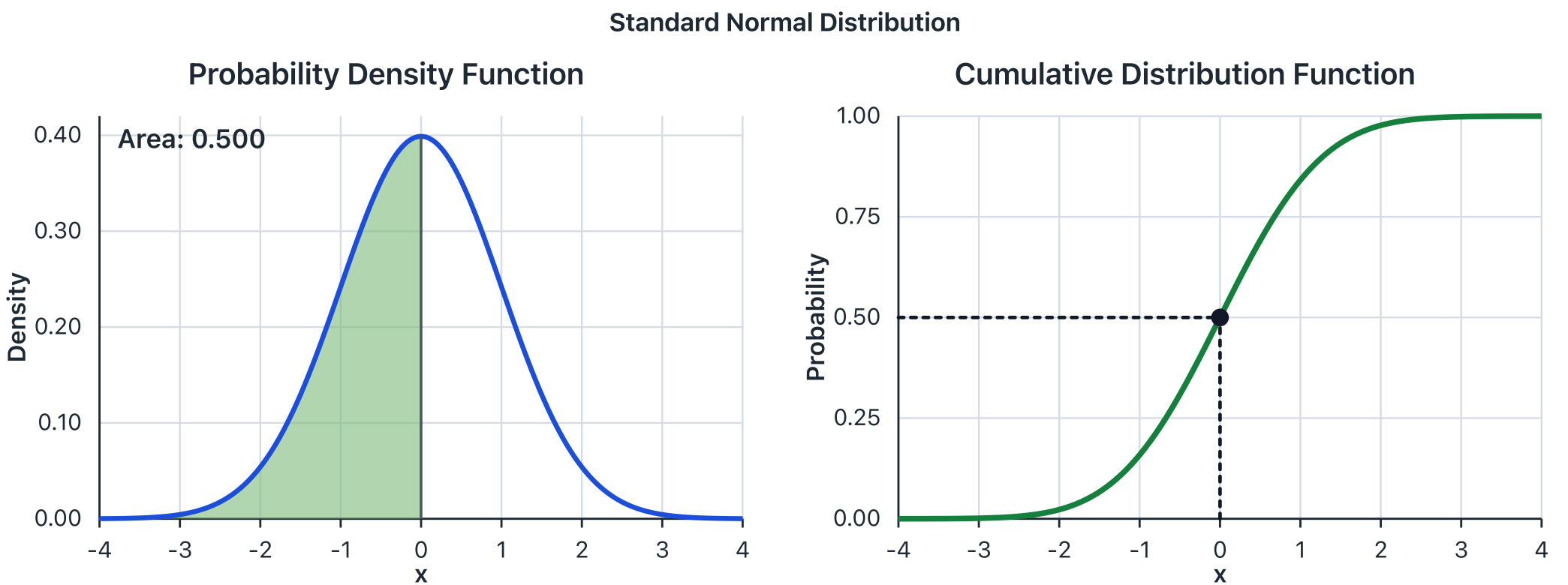
The standard normal

Example 1.2.19 — Standard Normal Distribution

The standard normal distribution, denoted by $N(0,1)$, has PDF ϕ and CDF Φ:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad \Phi(x) = \int_{-\infty}^{x} \phi(u)\, du.$$

Select a value for x: **0.00**



Support

Definition 1.2.20 — Support

For a random variable X with PMF/PDF f, the support of X is

$$\text{Supp}[X] = \{x \in \mathbb{R} : f(x) > 0\}.$$

- The values that can actually occur
  - Die roll: $\{1, \dots, 6\}$
  - standard uniform: $[0, 1]$
- What is the support of a standard normal?

Joint distributions

Definition 1.3.3 — Joint PMF

For discrete random variables X and Y, the joint PMF of X and Y is

$$f(x, y) = \Pr[X = x, Y = y], \ \forall x, y \in \mathbb{R}.$$

Example 1.3.5 — Flipping a Coin and Rolling a Die

$X = 0$ for heads (roll a 4-sided die), $X = 1$ for tails (roll a 6-sided die); Y = the roll. Then

$$f(x, y) = \begin{cases} \frac{1}{8} & : x = 0, y \in \{1, 2, 3, 4\} \\ \frac{1}{12} & : x = 1, y \in \{1, \dots, 6\} \\ 0 & : \text{otherwise.} \end{cases}$$

Marginal distributions

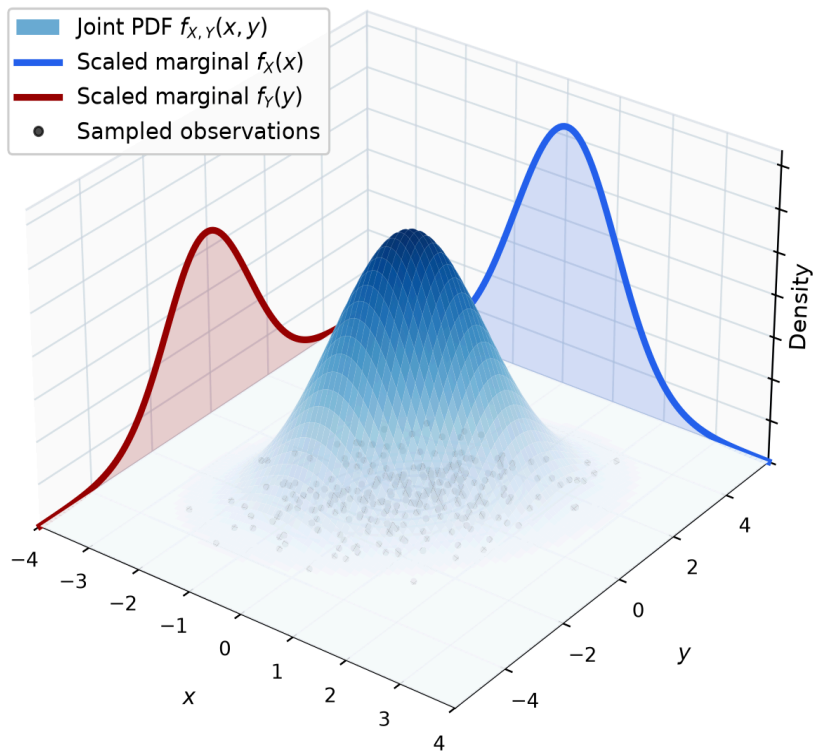
Theorem 1.3.6 — Marginal PMF

For discrete random variables X and Y with joint PMF f, the marginal PMF of Y is

$$f_Y(y) = \Pr[Y = y] = \sum_{x \in \text{Supp}[X]} f(x, y), \ \forall y \in \mathbb{R}.$$

- The distribution of Y alone, ignoring X — sum the joint over the other variable
- Follows from the Law of Total Probability

Joint/marginal distributions



$\mu = (0, 0), \quad \Sigma = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 2 \end{bmatrix}$. Marginal heights are scaled for visual comparison.

Conditional distributions

Definition 1.3.7 — Conditional PMF

For discrete random variables X and Y with joint PMF f, the conditional PMF of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \Pr[Y = y \mid X = x] = \frac{f(x, y)}{f_X(x)},$$

$\forall y \in \mathbb{R}$ and $\forall x \in \text{Supp}[X]$.

- The PMF of Y after "throwing out" all draws where $X \neq x$
- Same structure as conditional probability of events

Simulations

```
n = 100_000
X = (rng.random(n) < 0.5).astype(int)              # 1 = tails
Y = np.where(X == 0, rng.integers(1, 5, size=n),
                     rng.integers(1, 7, size=n))

print('joint    f(0,3)      ≈', ((X == 0) & (Y == 3)).mean())   # 1/8
print('marginal f_Y(3)      ≈', (Y == 3).mean())                # 5/24
print('cond.    f_X|Y(0|3)  ≈', (X[Y == 3] == 0).mean())        # 3/5
```

```
joint    f(0,3)      ≈ 0.12607
marginal f_Y(3)      ≈ 0.20858
cond.    f_X|Y(0|3)  ≈ 0.604420366286317
```

- Same generative process as the earlier Bayes example — now in random variable notation

Continuous versions

For jointly continuous r.v.s X and Y with joint PDF f:

Theorem 1.3.13 — Marginal PDF

The marginal PDF of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx, \ \forall y \in \mathbb{R}.$$

Definition 1.3.14 — Conditional PDF

The conditional PDF of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}, \ \forall y \in \mathbb{R} \text{ and } \forall x \in \text{Supp}[X].$$

Independence of random variables

Definition 1.3.16 — Independence of Random Variables

Let X and Y be either two discrete random variables with joint PMF f or two jointly continuous random variables with joint PDF f. Then X and Y are independent if, $\forall x, y \in \mathbb{R}$,

$$f(x, y) = f_X(x)\, f_Y(y).$$

- Written $X \perp\!\!\!\perp Y$
- Coin-and-die: *not* independent — $f(0, 3) = \frac{1}{8} \neq \frac{1}{2} \cdot \frac{5}{24}$

What independence buys us

Theorem 1.3.17 — Implications of Independence (Part I)

The following statements are equivalent:

- $X \perp\!\!\!\perp Y$
- $\forall x, y \in \mathbb{R},\ f(x, y) = f_X(x)\, f_Y(y)$
- $\forall x \in \mathbb{R}$ and $\forall y \in \operatorname{Supp}[Y]$, $f_{X|Y}(x \mid y) = f_X(x)$
- $\forall D, E \subseteq \mathbb{R}$, the events $\{X \in D\}$ and $\{Y \in E\}$ are independent
- For all functions g of X and h of Y, $g(X) \perp\!\!\!\perp h(Y)$

Random vectors

Definition 1.4.1 — Random Vector

A random vector (of length K) is a function $\mathbf{X} : \Omega \to \mathbb{R}^K$ such that, $\forall \omega \in \Omega$,

$$\mathbf{X}(\omega) = \left(X_{[1]}(\omega), X_{[2]}(\omega), \dots, X_{[K]}(\omega)\right),$$

where $\forall k \in \{1, 2, \dots, K\}$, $X_{[k]}$ is a random variable.

- Everything generalizes: joint/marginal/conditional PMFs, PDFs, CDFs
- Notation: bracketed subscripts $X_{[k]}$ for components; plain subscripts X_i reserved for repeated draws (Chapter 3)